



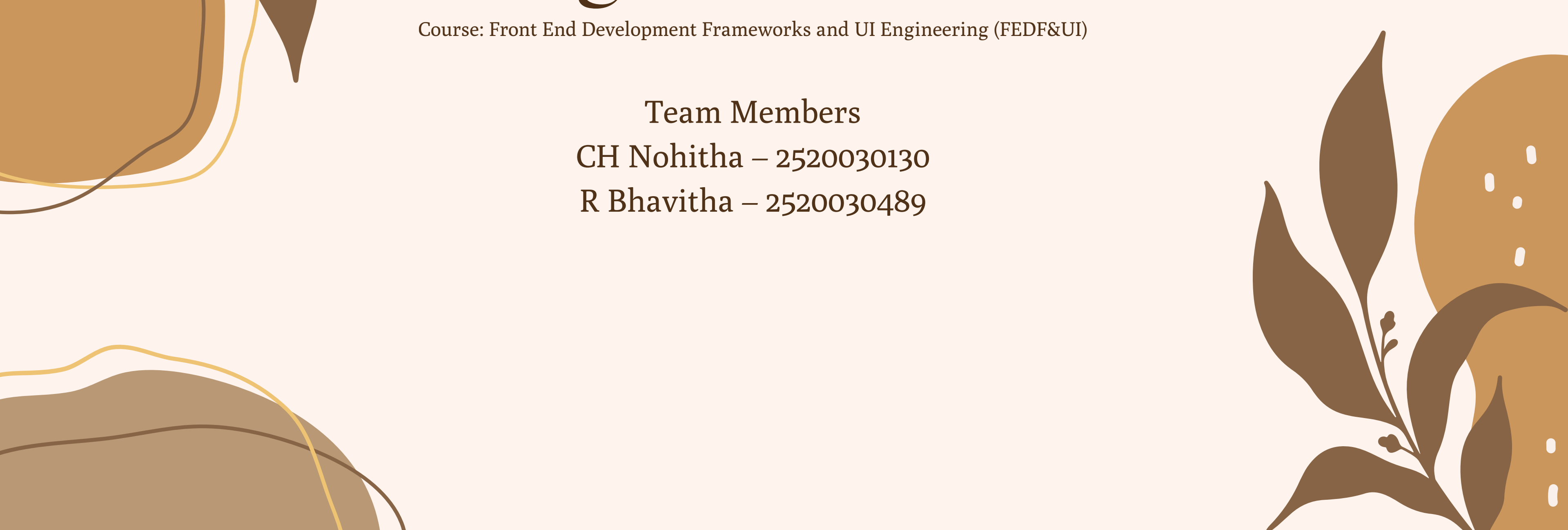
# RENTZY – Rental Management Portal

Course: Front End Development Frameworks and UI Engineering (FEDF&UI)

## Team Members

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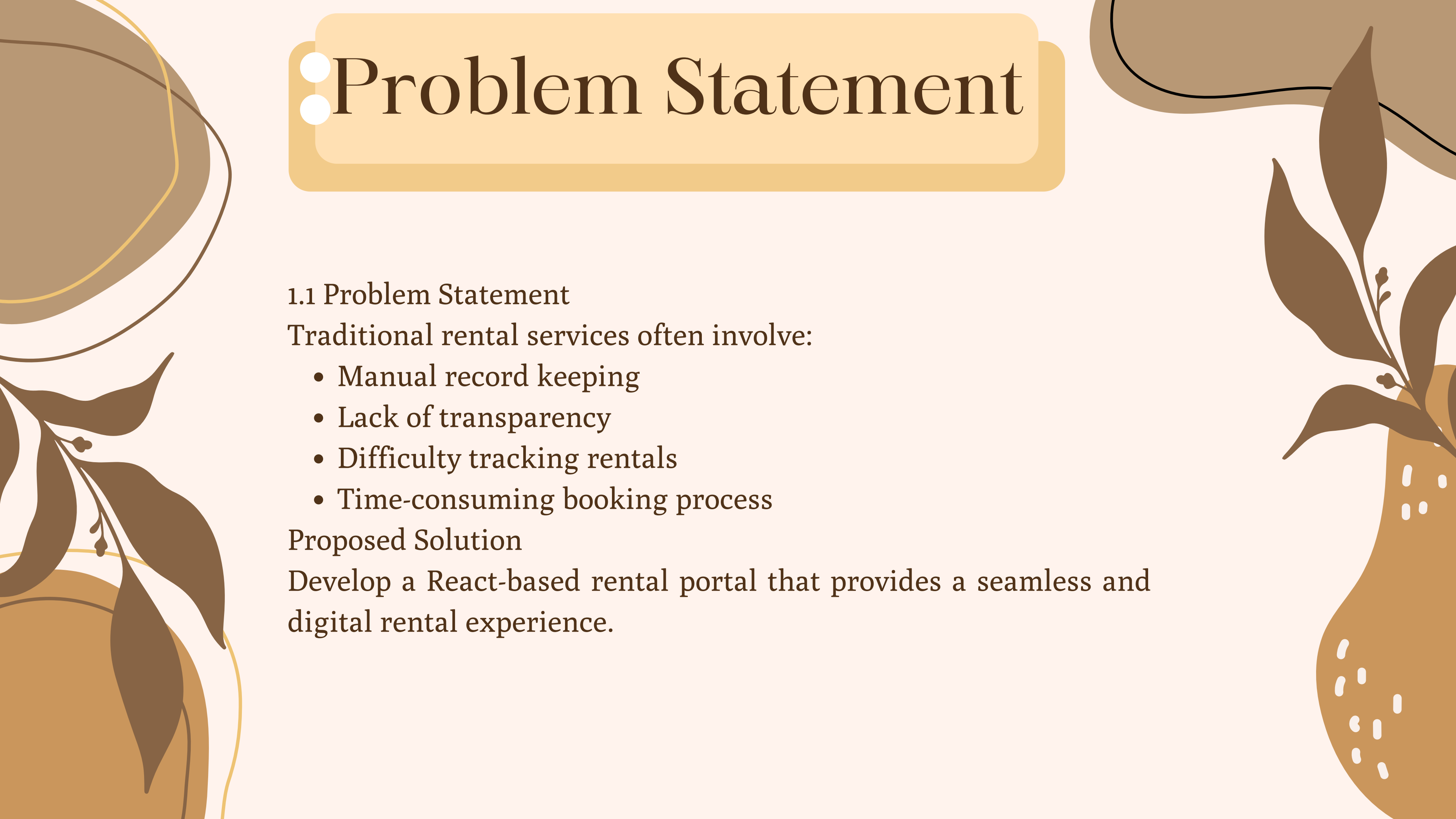


# Project Overview

Rentzy is a web-based rental management platform that allows users to browse, rent, and manage various products online. The system simplifies the rental process through a user-friendly interface and real-time order tracking.

## Modules:

- User Authentication
- Product Catalogue
- Rental Booking
- Order Management
- User Profile



# ●● Problem Statement

## 1.1 Problem Statement

Traditional rental services often involve:

- Manual record keeping
- Lack of transparency
- Difficulty tracking rentals
- Time-consuming booking process

## Proposed Solution

Develop a React-based rental portal that provides a seamless and digital rental experience.



# Objectives & Features

- 1.2 Objectives & Features

## Objectives

- Simplify rental booking
- Improve user experience
- Enable order tracking
- Maintain rental records digitally

## Features

- Login & Registration
- Product Browsing
- Rental Duration Selection
- Order Tracking
- User Profile Management



# System Requirements

## 2.1 Software & Tools Used

| Software     | Purpose              |
|--------------|----------------------|
| VS Code      | Development          |
| React        | Frontend Framework   |
| TypeScript   | Programming Language |
| Vite         | Build Tool           |
| Git & GitHub | Version Control      |
| Browser      | Testing              |

# Tech Stack

## 2.2 Tech Stack

### Frontend

- React.js
- TypeScript
- Tailwind CSS
- ShadCN UI Components

### Backend/API

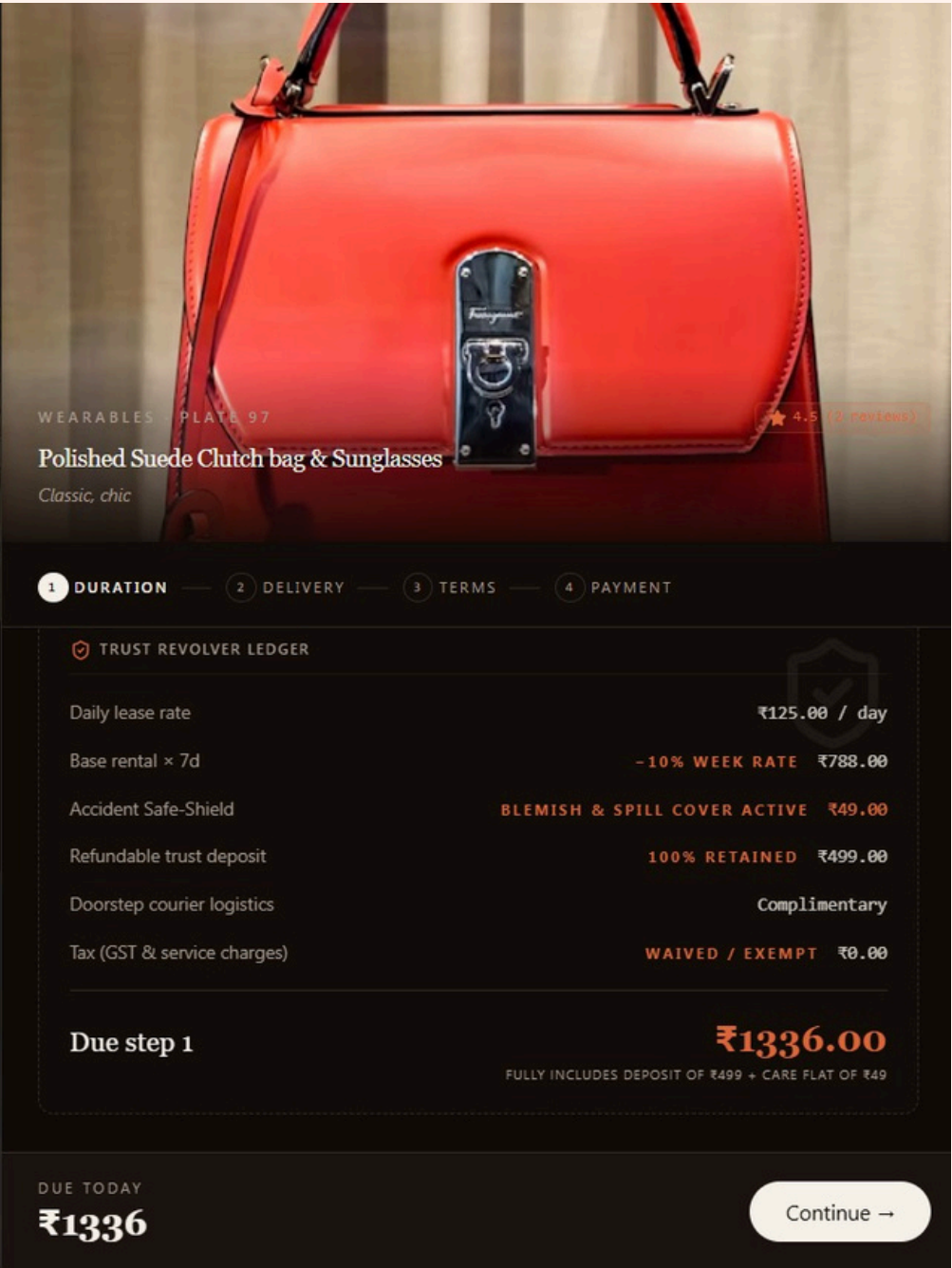
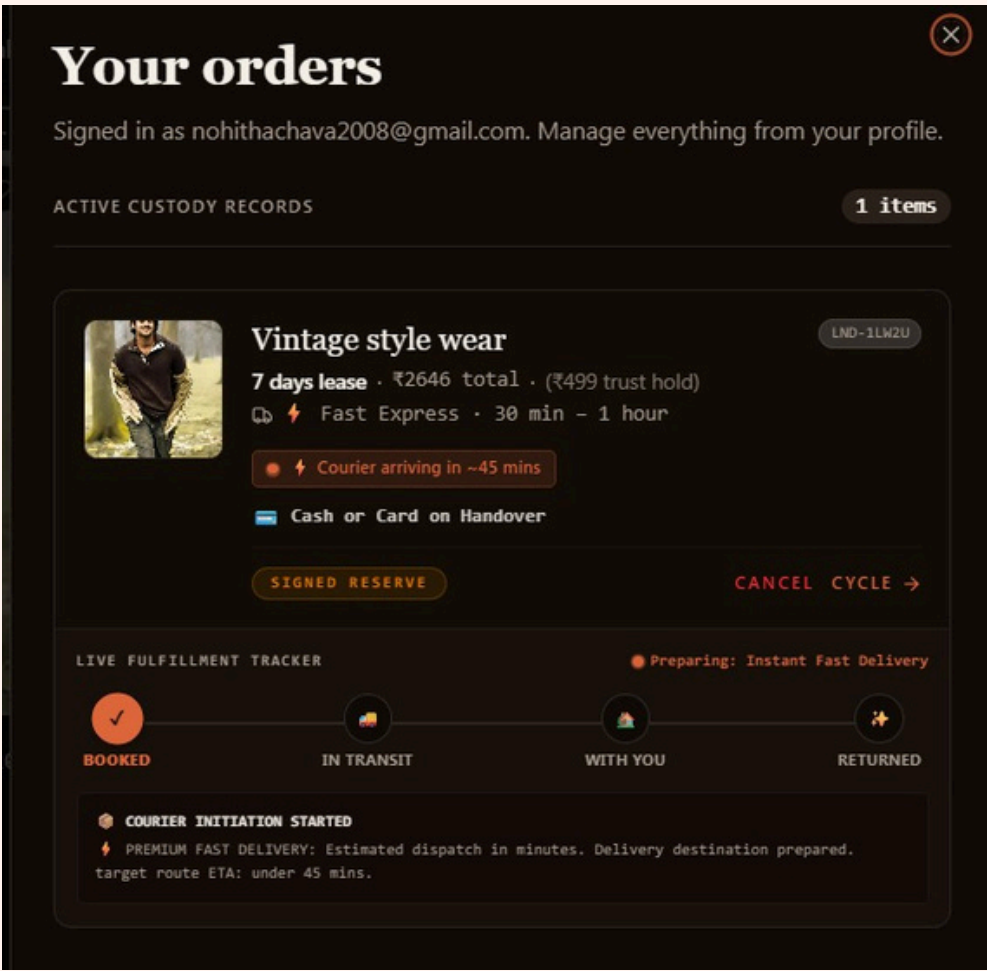
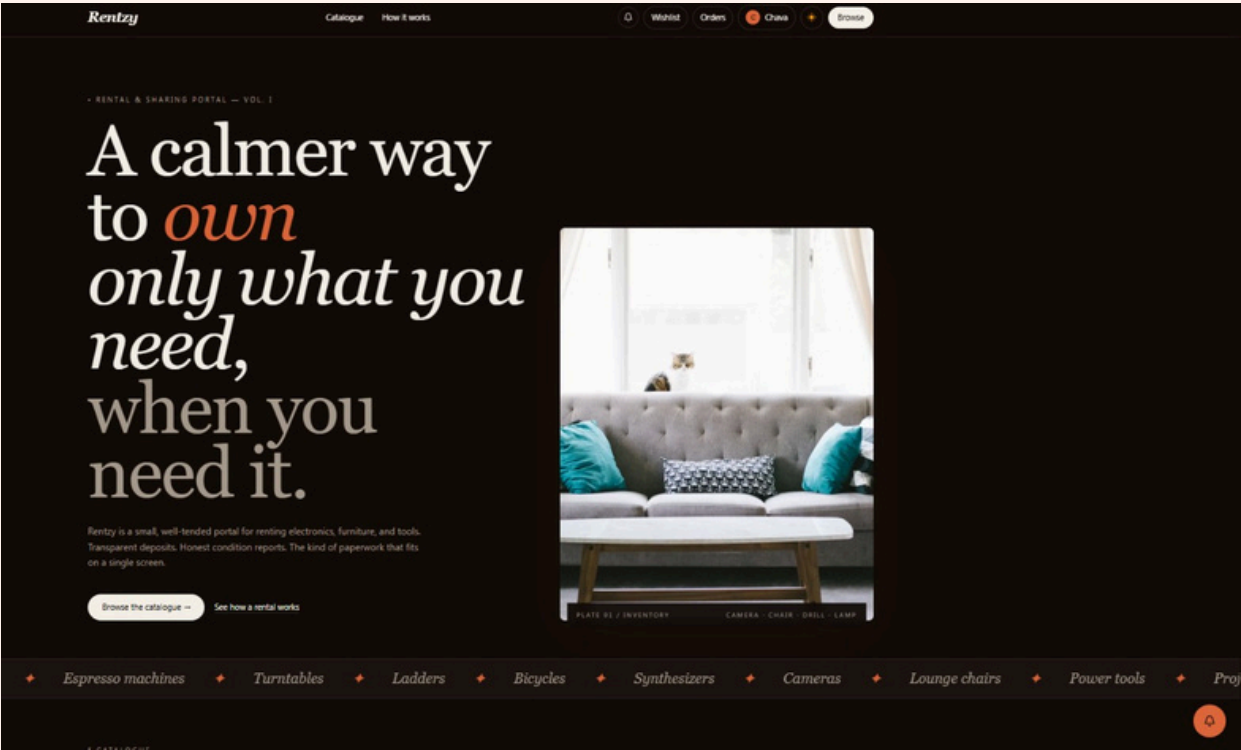
- Local Storage Based Data Management
- Authentication Module
- Rental Data Services

### Architecture

User → React UI → Business Logic → Data Storage



# App Layout



## 3.1 Application Layout

### Home Page

Login/Register

Product Catalogue

Rental Checkout

Profile Page

Order Tracking

# React Component Folder Structure

```
src/  
|  
├─ routes/  
|   ├─ index.tsx  
|   ├─ login.tsx  
|   └─ profile.tsx  
|  
├─ components/  
|   └─ ui/  
|  
├─ assets/  
|  
├─ lib/  
|   ├─ auth.ts  
|   └─ rental-data.ts  
|  
└─ main.tsx
```



# Main Application Logic

## 4.1 Main Application Logic User Authentication

`</>` TypeScript

```
const [email, setEmail] = useState("");  
const [password, setPassword] = useState("");  
  
const u = signIn(email, password);  
navigate({ to: "/profile" });
```

### Purpose

- Validates user credentials
- Redirects user after login

# React Hooks

## 4.2 Use of React Hooks

### useState

`</> TypeScript`

```
const [user, setUser] = useState(null);  
const [orders, setOrders] = useState([]);
```

### useEffect

`</> TypeScript`

```
useEffect(() => {  
  const u = currentUser();  
  setUser(u);  
}, []);
```

### Benefits

- State Management
- Dynamic UI Updates
- Lifecycle Handling

## Additional Code Snippets

### Rental Order Management

```
</> TypeScript  
  
setOrders(loadOrders(user.email));
```

### Statistics Calculation

```
</> TypeScript  
  
const totalSpent =  
orders.reduce((s, o) => s + o.total, 0);
```

### Navigation

```
</> TypeScript  
  
navigate({ to: "/profile" });
```

These snippets demonstrate:

- Data Retrieval
- Order Processing
- Page Navigation

# App Demo & Screenshots

[Sign in](#) / [Open account](#)

### Open an account.

Takes about twelve seconds. Stored on this device.

FULL NAME

Aarav Sharma

CITY

Hyderabad

EMAIL

aarav@sharma.in

PASSWORD

\*\*\*\*\*

Open account →

By continuing you accept the standard rental terms. — [Back to catalogue](#)

[Sign in](#) / [Open account](#)

### Welcome back.

Pick up where you left off — your reservations are waiting.

EMAIL

aarav@sharma.in

PASSWORD

\*\*\*\*\*

Sign in →

By continuing you accept the standard rental terms. — [Back to catalogue](#)

**Rentzy**

ACCOUNT

### A quieter *key* to the things you almost own.

One small account. Reservations, deposits, condition reports, and a tidy countdown of how many days are left on each rental.

V1.0.0 - LOCAL-FIRST

**Rentzy**

CatalogueHow it works

COMPUTING 4.5 1980/-  
Silver Pro Laptop, 14"  
Quiet, capable

COMPUTING 4 1420/-  
Tablet Pro & Pencil  
Light, tactile

DRONES 4.5 4780/-  
Compact Camera Dr...  
Quiet, aerial

DRONES 4.5 1430/-  
Cinematic FPV Racin...  
Aplis, aerobatic

DRONES 4.5 1420/-  
Miniature Whisper D...  
Quiet, pocket-sized

WEARABLES 4.5 1520/-  
Gold Dress Watch  
Quiet, formal

WEARABLES 4.5 1780/-  
Steel Dive Watch, 30...  
Heavy, formal

WEARABLES 4.5 1420/-  
Unisex Smart Watch  
Intuitive, high-tech

WEARABLES 4.5 1115/-  
Designer Pearl Accen...  
Timeless, luminous

WEARABLES 4.5 1125/-  
Polished Suede Clute...  
Classic, chic

APPAREL 4.5 1420/-  
Vintage style wear  
Warm, formal

APPAREL 4.5 1420/-  
Grey Blazer Paired w...  
Sharp, ceremonial

**Rentzy**

CatalogueSign out

ACCOUNT

## Hello, *Nohitha*.

Nothing out at the moment. The catalogue is open whenever you are.

NOHITHACHAVA2008@GMAIL.COM · MEMBER SINCE JUN 2026

ACTIVE

0

COMPLETED

0

LIFETIME SPEND

₹0

DEPOSIT ON HOLD

₹0

WORKSPACE SWITCHBOARD CONSOLE

Customer HubAdministrative Desk

PROFILE

EDIT

NAME

Nohitha

EMAIL

nohithachava2008@gmail.com

CITY

Hyderabad

NOTIFICATIONS

Delivery reminders

Return-day nudge

New arrivals (weekly)

### Your Wishlist

0 saved

Your wishlist is empty.

Click the heart icon on any asset in the catalogue to save it here for later.

BROWSE OBJECTS

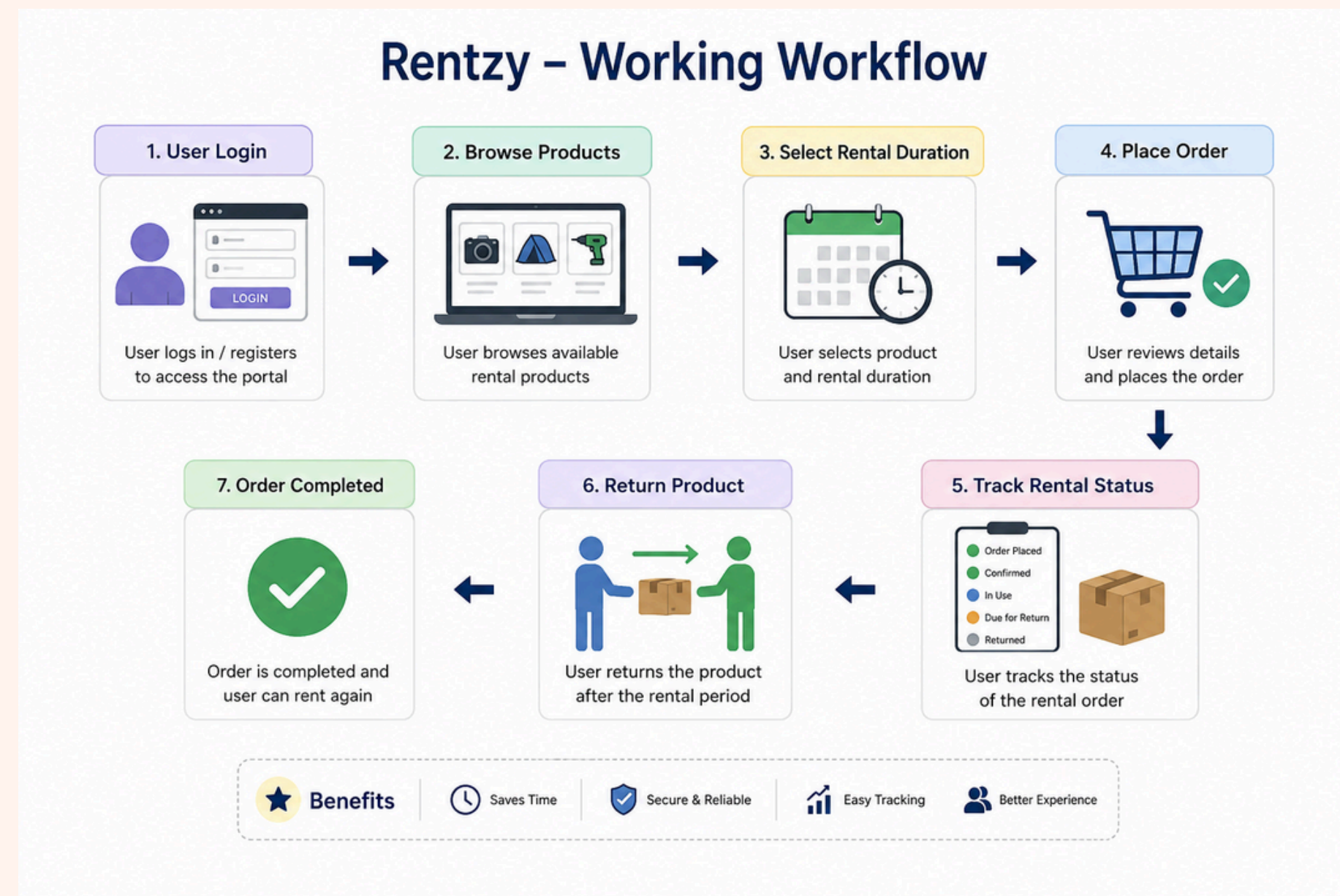
### Active rentals

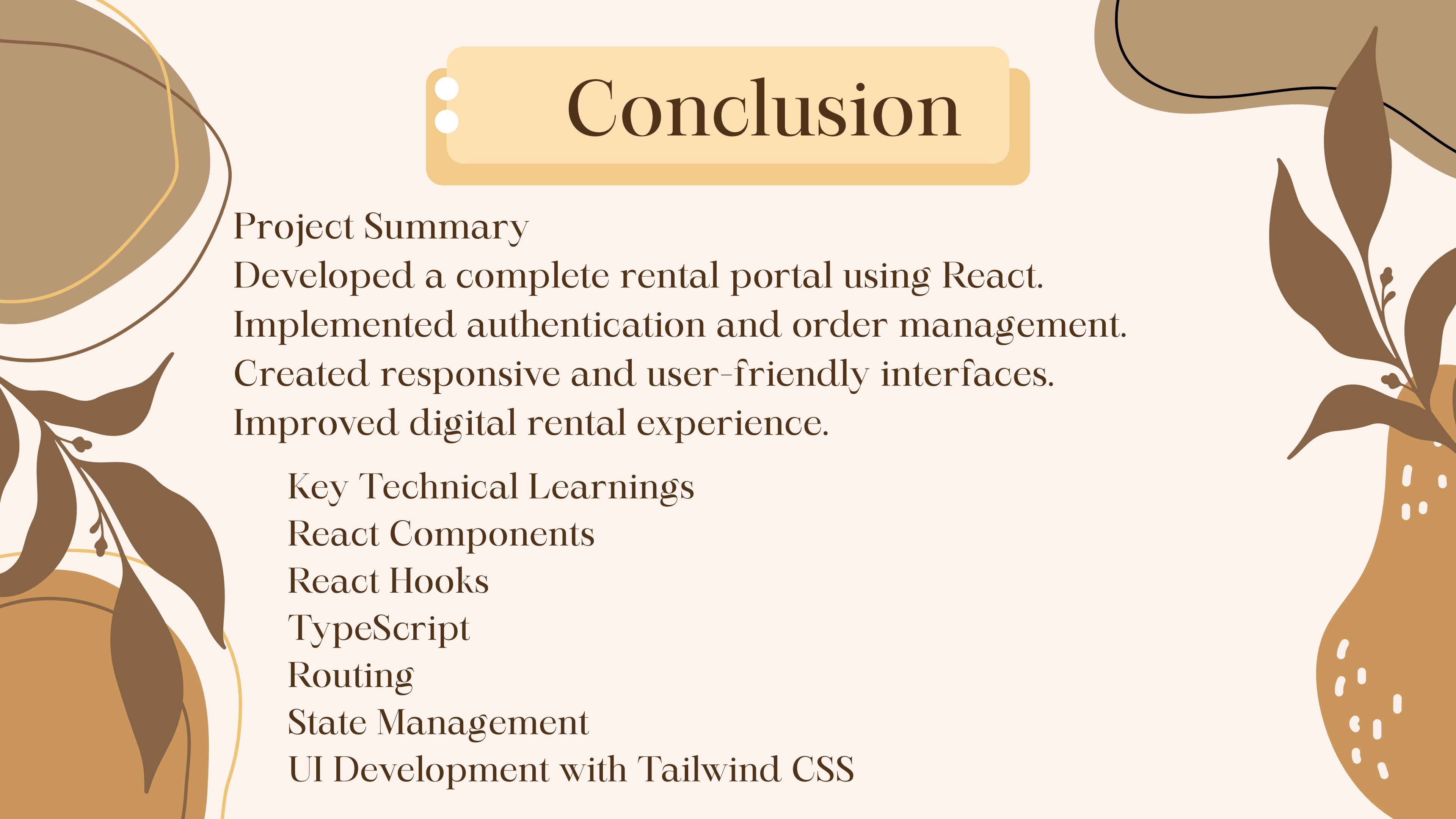
0 open

CLEAR ORDER HISTORY

# Workflow Example

## 5.2 Working Workflow





# Conclusion

## Project Summary

Developed a complete rental portal using React.  
Implemented authentication and order management.  
Created responsive and user-friendly interfaces.  
Improved digital rental experience.

## Key Technical Learnings

React Components

React Hooks

TypeScript

Routing

State Management

UI Development with Tailwind CSS



Thank You